

Ian Drosos

	Member of Technical Staff Trent AI Cognitive Science Ph.D.	Updated: August 17, 2026 linkedin.com/in/ian-drosos/ iandrosos.me
RESEARCH INTERESTS	Human-computer interaction; human-centered AI; product design and UX engineering for agentic systems and security; generative and dynamic interfaces; human oversight, AI steering, and decision support; tools for developers and knowledge workers.	
EDUCATION	University of California, San Diego Ph.D. in Cognitive Science Dissertation: <i>Synthesizing Transparent and Inspectable Technical Workflows</i> , Advisor: Philip Guo	2017 – 2022
	North Carolina State University M.S. in Computer Science Thesis: <i>HappyFace: Identifying and Predicting Frustrating Learning Obstacles at Scale</i> , Advisor: Chris Parnin	2015 – 2017
	Southern Polytechnic State University B.S. in Computer Science	2007 – 2011
EXPERIENCE	Trent AI, London, UK <i>Member of Technical Staff</i> Lead UX design engineering across Trent AI's agentic security products as the primary designer and a hands-on 0-to-1 product engineer, taking the core experience from early concepts through production. Design and implement generative UX patterns for AI agents. Drive product redesigns that reduce cognitive load, translating complex security analysis into clear and actionable workflows for developers and non-experts. Translate HCI research on AI steering, critical thinking, and human oversight into shipped product experiences for high-stakes security decisions.	2025 –
	Microsoft Research, Cambridge, UK <i>Researcher</i> HCI + AI research of the design of intelligent systems on the Tools for Thought project . I study how AI can help users critically think and have greater control over AI-generated output during data analysis and programming tasks through HCI research of professional, end-user, and learner workflows. [C.8-15; C.X1-2; W.1-2; W.X1]	2022 – 2025
	UCSD – The Design Lab, La Jolla, CA <i>Researcher – Ph.D. Candidate</i> HCI research in providing better experiences for data scientists, programmers, content creators, and learners. Published six papers and won two paper awards. [C.2-7]	2017 – 2022

UCSD, La Jolla, CA*Instructor*

2018 – 2022

- HCI Portfolio Design Studio (COGS121)

Teaching Assistant

- Interaction Design (COGS120/CSE170)

- Human-Computer Interaction Programming Studio (COGS121)

- HCI Portfolio Design Studio (COGS121)

- Data-Driven UX/Product Design (COGS127)

Autodesk, San Rafael, CA*Intern – User Interface Research*

01/2021 – 04/2021

Investigated barriers to providing expert (human) help to questions about feature-rich software like Autodesk Fusion 360. Designed, implemented, and deployed a custom survey prototype to collect feedback from experts (n=28). Paper in submission. [C.16].

Microsoft, Redmond, WA*Research Intern – Program Synthesis*

07/2018 – 12/2018

Designed, implemented, and evaluated (n=12) a prototype called Wrex for generating readable Python code through program synthesis within Jupyter notebooks using JavaScript and Python. Published the results at CHI2020, which won Best Paper. [C.4]

Verizon, Alpharetta, GA*Member Technical Staff I & II – Systems Engineering*

2011 – 2015

Full-stack software engineer developing enterprise systems using Java, PL/SQL, JavaScript, and HTML.

PUBLICATIONS

(C)onference, (J)ournal, and (W)orkshop.

C.16 Ian Drosos, Jo Vermeulen, George Fitzmaurice, Justin Matejka. 2026. Nanomen-
toring: Investigating How Quickly People Can Help People Learn Feature-Rich
Software. In Proceedings of the 14th Nordic Conference on Human-Computer
Interaction (NordiCHI '26). [\[Link\]](#)

W.3 Advait Sarkar and Ian Drosos. 2025. Vibe coding: programming through con-
versation with artificial intelligence. In the Workshop of the Psychology of Pro-
gramming Interest Group (PPIG 2025). [\[Link\]](#)

C.15 Ian Drosos, Jack Williams, Advait Sarkar, Nicholas Wilson, Sean Rintel, and
Payod Panda. 2025. Dynamic Prompt Middleware: Contextual Prompt Refine-
ment Controls for Comprehension Tasks. In Proceedings of the 4th Annual Sym-
posium on Human-Computer Interaction for Work (CHIWORK 2025). (Patent
filed). [\[Link\]](#) ***2nd Place / 1,064 projects in internal Hackathon (Every-
day AI Executive Challenge)**

C.14 Hao-Ping (Hank) Lee, Advait Sarkar, Lev Tankelevitch, Ian Drosos, Sean Rin-
tel, Richard Banks, and Nicholas Wilson. 2025. The Impact of Generative AI
on Critical Thinking: Self-Reported Reductions in Cognitive Effort and Confi-
dence Effects From a Survey of Knowledge Workers. In Proceedings of the CHI
Conference on Human Factors in Computing Systems (CHI 2025). [\[Link\]](#)

- C.13 Bhuvanashree Murugadoss, Christian Poelitz, Ian Drosos, Vu Le, Nick McKenna, Carina Negreanu, Chris Parnin, and Advait Sarkar. 2025. Evaluating the Evaluator: Measuring LLMs’ Adherence to Task Evaluation Instructions. In Proceedings of the AAAI Conference on Artificial Intelligence (AAAI 2025). [\[Link\]](#)
- C.12 Majeed Kazemitabaar, Jack Williams, Ian Drosos, Tovi Grossman, Austin Henley, Carina Negreanu, and Advait Sarkar. 2024. Improving Steering and Verification in AI-Assisted Data Analysis with Interactive Task Decomposition. In Proceedings of The ACM Symposium on User Interface Software and Technology (UIST 2024). [\[Link\]](#)
- C.11 Advait Sarkar, Xiaotong (Tone) Xu, Neil Toronto, Ian Drosos, and Christian Poelitz. 2024. When Copilot Becomes Autopilot: Generative AI’s Critical Risk to Knowledge Work and a Critical Solution. The European Spreadsheet Risks Interest Group Conference (EuSpRIG 2024). [\[Link\]](#)
- C.10 Ian Drosos, Advait Sarkar, Xiaotong (Tone) Xu, Carina Negreanu, Sean Rintel, and Lev Tankelevitch. 2024. “It’s like a rubber duck that talks back”: Understanding Generative AI-Assisted Data Analysis Workflows through a Participatory Prompting Study. In Proceedings of the Symposium on Human-Computer Interaction for Work. (CHIWORK 2024). [\[Link\]](#)
- W.2 Andrew D. Gordon, Carina Negreanu, José Cambronero, Rasika Mudumbai Chakravarthy, Ian Drosos, Hao Fang, Bhaskar Mitra, Hannah Richardson, Advait Sarkar, Stephanie Simmons, Jack Williams, Ben Zorn. 2024. Co-audit: tools to help humans double-check AI-generated content. In the Workshop on Evaluation and Usability of Programming Languages and Tools (PLATEAU 2024). [\[Link\]](#)
- W.1 Advait Sarkar, Ian Drosos, Rob DeLine, Andrew D. Gordon, Carina Negreanu, Sean Rintel, Jack Williams, and Ben Zorn. 2023. Participatory prompting: a user-centric research method for eliciting AI assistance opportunities in knowledge workflows. In the Workshop of the Psychology of Programming Interest Group (PPIG 2023). [\[Link\]](#)
- C.9 Ian Drosos, Nick Wilson, Andrew D. Gordon, Sruti Ragavan, and Jack Williams. 2023. FxD: a functional debugger for dysfunctional spreadsheets. In Proceedings of the Symposium on Visual Languages and Human-Centric Computing (VL/HCC 2023). (Patent filed). [\[Link\]](#). *FxD is now part of [Excel Labs!](#)*
Best Paper, Honorable Mention Award
- C.8 Kasra Ferdowsi, Jack Williams, Ian Drosos, Andrew D. Gordon, Carina Negreanu, Advait Sarkar, Benjamin Zorn. 2023. ColDeco: An End User Spreadsheet Inspection Tool for AI-Generated Code. In Proceedings of the Symposium on Visual Languages and Human-Centric Computing (VL/HCC 2023). (Patent filed). [\[Link\]](#)
- C.7 Ian Drosos and Philip Guo. 2022. The Design Space of Livestreaming Equipment Setups: Tradeoffs, Challenges, and Opportunities. In Proceedings of the ACM Designing Interactive Systems Conference (DIS 2022). [\[Link\]](#)

	C.6 Ian Drosos and Philip Guo. 2021. Streamers Teaching Programming, Art, and Gaming: Cognitive Apprenticeship, Serendipitous Teachable Moments, and Tacit Expert Knowledge. In Proceedings of the Symposium on Visual Languages and Human-Centric Computing, short paper (VL/HCC 2021). [Link] *Best Short Paper, Honorable Mention Award*
	C.5 Sam Lau, Ian Drosos, Julia Markel and Philip Guo. 2020. The Design Space of Computational Notebooks: An Analysis of 60 Systems in Academia and Industry. In Proceedings of the Symposium on Visual Languages and Human-Centric Computing (VL/HCC 2020). [Link]
	C.4 Ian Drosos, Titus Barik, Philip Guo, Robert DeLine, and Sumit Gulwani. 2020. Wrex: A Unified Programming-By-Example Interaction for Synthesizing Readable Code for Data Scientists. In Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI 2020). [Link] *Best Paper Award*
	C.3 Adam Rule, Ian Drosos, Aurélien Tabard, and James D. Hollan. 2018. Aiding Collaborative Reuse of Computational Notebooks with Annotated Cell Folding. In Proceedings of the ACM on Human-Computer Interaction (CSCW 2018). [Link]
	C.2 René Just, Chris Parnin, Ian Drosos, and Michael D. Ernst. 2018. Comparing developer-provided to user-provided tests for fault localization and automated program repair. In Proceedings of the ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2018). [Link]
	C.1 Ian Drosos, Philip Guo, and Chris Parnin. 2017. HappyFace: Identifying and Predicting Frustrating Obstacles for Learning Programming at Scale. In Proceedings of the Symposium on Visual Languages and Human-Centric Computing (VL/HCC 2017). [Link]
IN SUBMISSION	C.X2 Ian Drosos, Advait Sarkar, Xiaotong (Tone) Xu, and Neil Toronto. 2025. “It makes you think”: Provocations Restore Critical Thinking During AI-Assisted Tasks. (In review). [Link] C.X1 Ian Drosos, Advait Sarkar, and Andrew D. Gordon. 2023. “My toxic trait is thinking I’ll remember this”: Gaps in the learner experience of video tutorials for feature-rich software. (In review). [Link]
PATENTS AND AWARDS	<i>Patents</i> - Rendering gui components for refining queries to generative machine learning models. Jack Williams, Ian Drosos, Advait Sarkar, Nicholas Wilson. 2026. App US/18934192. [C.15] - Techniques for formula debugging. Advait Sarkar, Sruti Srinivasa Ragavan, Jack Williams, Ian Drosos, Nicholas Wilson, Irena Berezovsky, Lev Solodkin, Andrew Gordon. 2025. App US/18401217. [C.9] - Range preview with elisions. Advait Sarkar, Sruti Srinivasa Ragavan, Jack Williams, Ian Drosos, Nicholas Wilson, Irena Berezovsky, Lev Solodkin, Andrew Gordon. 2025. App US/18401240. [C.9] - Generation of multi-dimensional array of intermediate values in multiple code executions. Benjamin Zorn, Kasra Ferdowsi, Jack Williams, Carina Negreanu, Andrew Gordon, Advait Sarkar, Ian Drosos. 2024. App US/18154741. [C.8]

Paper Awards

- VL/HCC 2023. Best Paper, Honorable Mention Award. [C.9]
- VL/HCC 2021. Best Short Paper, Honorable Mention Award. [C.6]
- CHI 2020. Best Paper Award. [C.4]

Hackathons

- Microsoft Global Hackathon 2024 Executive Challenge Winner (2nd place / 1,064 projects in the ‘Everyday AI’ Challenge). Promptly: Dynamically generated UI for AI.
- Microsoft Global Hackathon 2023 Executive Challenge Winner (2nd place / 182 projects in the ‘Change the Game’ Challenge). Game Development Copilot: Making a Dream Game.

SKILLS

Product & UX leadership: 0-to-1 product design, UX design engineering, interaction design, prototyping, and complex-workflow simplification

AI & security UX: agentic AI, generative and dynamic interfaces, conversational UX, AI steering, human-in-the-loop controls, high-stakes decision support, and security-assessment workflows

Research & evaluation: mixed-methods HCI, interviews, thematic analysis, content analysis, surveys, experiments, comparative tool studies, and design probes

Design engineering: Figma, high-fidelity prototyping, code prototyping, React, TypeScript, and JavaScript

Software engineering & analysis: Python, Java, SQL, and R

PROGRAMMING
LANGUAGES

JavaScript (React/Typescript), Python, R, Java, SQL

SERVICE

Program Committee

- Intelligent User Interfaces 2025-2026
- Learning@Scale 2023-2024
- VL/HCC 2023-2024

Reviewer

- CHI, VL/HCC, UIST

INVITED TALKS

Rising Stars Applied Research Talk Series - Designing Better Human-AI Workflows: Dynamic UX, Provocations, and Critical Engagement
Microsoft (E+D Applied Research), February 2025

LLM Forum - Understanding Generative AI-Assisted Data Analysis Workflows
European Bioinformatics Institute, September 2024

Learning programming in the era of LLMs
Google, January 2024

MENTORSHIP

Hank Lee, *Microsoft Research intern*, Summer 2024
Bhuvanashree Murugadoss, *Microsoft Research Fellow*, 2023-2024
Xiaotong (Tone) Xu, *Microsoft Research intern*, Summer 2023
Majeed Kazemitabaar, *Microsoft Research intern*, Summer 2023
Kasra Ferdowsi, *Microsoft Research intern*, Summer 2022